

Trigonometry Complete Mastery

Foundation (Class 9–10)

Introduction to Trigonometry

Trigonometric Identities

$\sin^2\theta + \cos^2\theta = 1$

$1 + \tan^2 \theta = \sec^2 \theta$

$1 + \cot^2 \theta = \csc^2 \theta$

$\sqrt{\frac{a^2+b^2}{c}}$

Right-angled triangle:
hypotenuse, base, perpendicular

Six trigonometric ratios (sin, cos,
tan, cosec, sec, cot)

Reciprocal and quotient relations

Ratios of 0°, 30°, 45°, 60°, 90°

Advanced notation

Inverse functions and powers

Further formula checks

$\overline{AB} + \underline{x} + \boxed{\pi r^2}$

$\frac{\sum_{i=1}^n x_i}{n}$

$\left(\frac{1}{1+x}\right)^2$

$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$\sin^{-1}(x), \cos^{-1}(x), \tan^{-1}(x)$

$x^{-2}, 10^{-3}, a_n, x_1, \text{H}_2\text{O}$

$\theta, \pi, \infty, \in, \rightarrow, \Leftrightarrow, \pm, -$

Superscripts $^{0123456789+-=()}$

Subscripts $_{0123456789+-=()}$